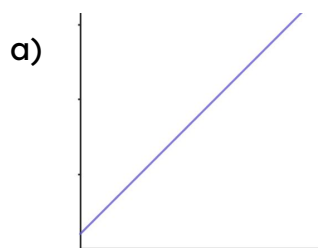




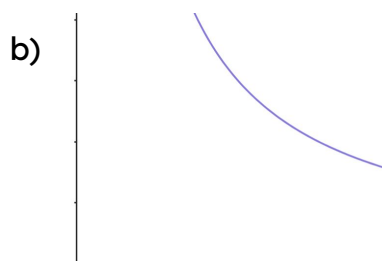
Inverse proportion in context

Task A: Graphs of inverse proportion

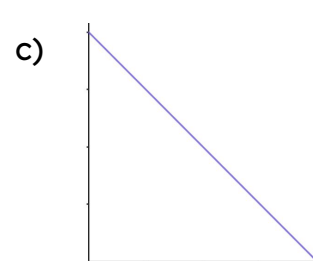
1) Label each of the graphs as direct proportion, **inverse proportion** or neither.



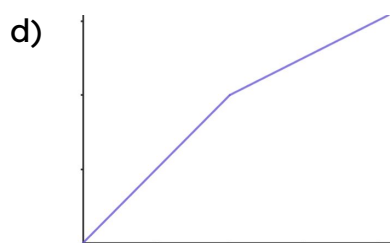
neither



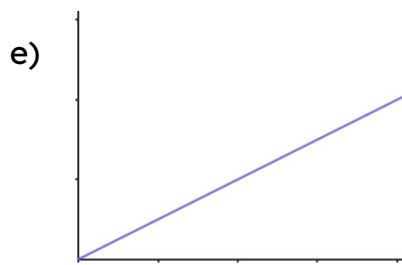
inverse proportion



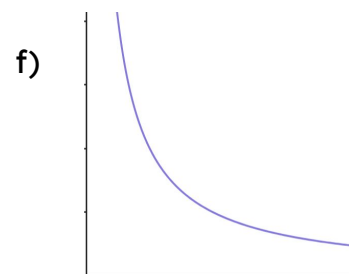
neither



neither



direct proportion



inverse proportion

2) Decide if the following situations are directly or **inversely proportional**:

- a) The number of taps and the time taken to fill a bath. *Inverse*
- b) The number of Euros exchanged and the number of pounds. *Direct*
- c) The number of boxes of tiles bought and the cost. *Direct*
- d) The number of days to dig a hole and the number of days it takes. *Inverse*
- e) The time taken to get to Edinburgh and the speed of the car. *Inverse*



Task B: Using inverse proportion to find missing values

- 1) The following tables show variables A & B which are **inversely proportional**. Find the missing value.

a)

A	B
6	20
12	10

b)

A	B
3	65
15	13

c)

A	B
16	24
2	192

d)

A	B
20	16
5	64

e)

A	B
12	8.75
5	21

f)

A	B
27	24
8	81

- 2) Use a ratio table to answer the following:

- a) It takes 4 people 2 days to paint a house. How long will it 8 people? *1 day*
- b) It takes 10 factory workers 4 hours to pack a lorry full of boxes. 2 people are on holiday. How long will it take to fill the lorry now? *5 hours*
- c) It takes 3 days to fill a pond if 5 taps are used. How long will it take if 4 taps are used. Give your answer in hours and minutes. *3 hours 45 minutes*
- d) 4 machines work all day every day. The 7y make 1 million items in 9 days. The company buys 1 new machine. How long will it take to make 1 million items now? Give your answer in days and hours. *days 12 mins*

